# Polar Differentiation

Questions 1–22

**Question 1.** For each of the following equations which are given in polar form, find $y$ and $x$ in terms of $\sin\theta$ and $\cos\theta$:

a) $r = 4\cos\theta$
b) $r = 8\sin\theta$
c) $r = 3(1+\cos\theta)$
d) $r = 6(1-\sin\theta)$
e) $r = 5\sin 2\theta$

**Question 2.** For each of the following, find $\dfrac{dy}{d\theta}$ giving your answer in terms of $\sin\theta$ and $\cos\theta$:

a) $y = 3\cos\theta + 3\cos^2\theta$
b) $y = 5\sin\theta - 5\sin^2\theta$
c) $y = 7\cos\theta - 7\cos^2\theta$
d) $y = -2\sin\theta - 2\sin^2\theta$
e) $y = 8\cos^3\theta - 4\cos\theta$
f) $y = 6\sin^2\theta\cos\theta$

**Question 3.** A polar curve is given by $r = 2(1 + \sin\theta)$ where $r > 0$ and $-\pi < \theta \leq \pi$.

a) Show that $\dfrac{dy}{d\theta} = 2\cos\theta + 2\sin 2\theta$.

b) Show that $\dfrac{dx}{d\theta} = -2\sin\theta + 2\cos 2\theta$.

c) Determine the values of $\theta$ for which $\dfrac{dy}{d\theta} = 0$.

d) Determine the values of $\theta$ for which $\dfrac{dx}{d\theta} = 0$.

e) Hence, without referring to the graph of $r = 2(1 + \sin\theta)$, identify where the curve has a cusp point.

**Question 4.** Each of the following polar curves are defined for $r > 0$ and $-\pi < \theta \leq \pi$. In each case, find the equation of the tangent lines to the curve which are parallel to the initial line $\theta = 0$:

a) $r = 2(1 + \cos\theta)$

b) $r = 3(1 - \sin\theta)$

c) $r = 4\sin 2\theta$

d) $r = a(1 + \sin\theta)$ for constant $a$

e) $r = a(1 - \cos\theta)$ for constant $a$

f) $r = a\cos 2\theta$ for constant $a$

g) $r = e^{\theta}$

**Question 5.** Each of the following polar curves are defined for $r > 0$ and $-\pi < \theta \leq \pi$. In each case, find the equation of the tangent lines to the curve which are perpendicular to the initial line $\theta = 0$:

a) $r = 4 - 4\cos\theta$
b) $r = 2 + 2\sin\theta$
c) $r = 2\cos 2\theta$
d) $r = a - a\sin\theta$ for constant $a$
e) $r = a + a\cos\theta$ for constant $a$
f) $r = a\sin 2\theta$ for constant $a$
g) $r = e^{\theta}$
h) $r = 1 + \tan\theta$

**Question 6.** A polar curve is given by $r = 2 + \sqrt{3}\cos\theta$, $0 \leq \theta < 2\pi$. A point $P$ where $0 < \phi < \dfrac{\pi}{2}$ is on the curve, and the tangent to the curve is parallel to the initial line.

a) Show that $\cos\phi = \dfrac{-1+\sqrt{7}}{2\sqrt{3}}$.
b) Hence, find the value of $r$ at the point $P$.

**Question 7.** A polar curve is given by $r = e^{2\theta}$, $0 \leq \theta \leq \pi$.

a) Sketch the curve.

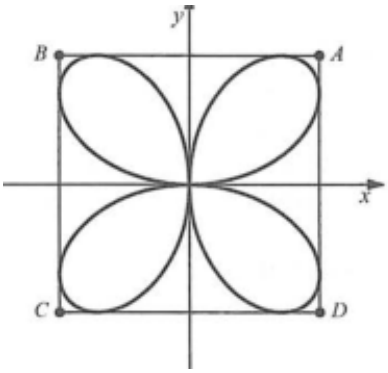
b) State the distance between the pole and the point furthest away from the pole.

Find the distance between the pole and each of the following points on the curve:

a) The point on the tangent line which is perpendicular to the initial line.

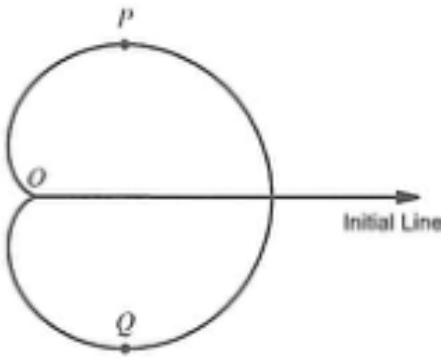
b) The point on the tangent line which is parallel to the initial line.

**Question 8.** A curve is given by $r = \frac{1}{2}\sin 2\theta$, $0 \leq \theta < 2\pi$, and is bounded by the square $ABCD$ as shown in the diagram below.




Calculate the exact area of the square $ABCD$.

**Question 9.** A polar curve is given by $r = a(1 + \cos\theta)$, $0 \leq \theta < 2\pi$, where $a$ is a positive constant.

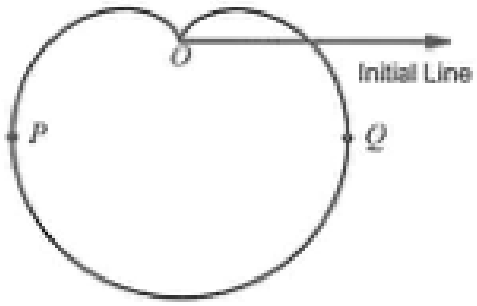



The points $P$ and $Q$ are where the tangent lines are parallel to the initial line. The distance between $P$ and $Q$ is known to be 12 units.

a) Determine the polar coordinates $(r, \theta)$ of the points $P$ and $Q$.

b) Find the value of $a$ giving your answer in exact form.

**Question 10.** A polar curve is given by $r = a(1 - \sin\theta)$, $0 \leq \theta < 2\pi$, where $a$ is a constant.




The points $P$ and $Q$ are where the tangent lines are perpendicular to the initial line. The distance between $P$ and $Q$ is known to be 5.

a) Determine the polar coordinates $(r, \theta)$ of the points $P$ and $Q$.

b) Find the value of $a$ giving your answer in exact form.

**Question 11.** A polar curve is given by $r = k - 3\cos\theta$, $0 \leq \theta < 2\pi$, where $k > 3$. It is known that there are exactly four points on the curve such that the tangent is perpendicular to the initial line.

a) Show that the range of possible values for $k$ is $3 < k < 6$.

b) Sketch $r = 5 - 3\cos\theta$ where $0 \leq \theta < 2\pi$. You should label any points at which it crosses the axes.

**Question 12.** A polar curve is given by $r = k - 4\sin\theta$, $0 \leq \theta < 2\pi$, where $k > 4$. It is given that there are exactly four points on the curve such that the tangent is parallel to the initial line.

a) Show that the range of possible values for $k$ is $4 < k < 8$.

b) Sketch $r = 6 - 4\sin\theta$ where $0 \leq \theta < 2\pi$. You should label any points at which it crosses the axes.

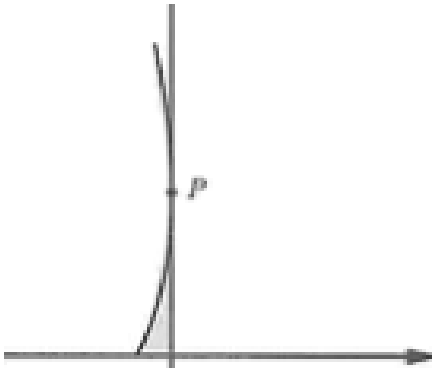
**Question 13.** A polar curve $C$ is given by $r = 4(1 - \sin\theta)$, $0 \leq \theta < \pi$.

a) Determine the horizontal line $L$ which is a tangent to the curve.

b) Sketch the curve $C$ and $L$ on the same axes.

c) Find the area enclosed by the curve $C$ and the line segment $L$, giving your answer in exact form.

**Question 14.** A polar curve $C$ is given by $r = 2 + \tan\theta$, $0 \leq \theta \leq \frac{\pi}{4}$.



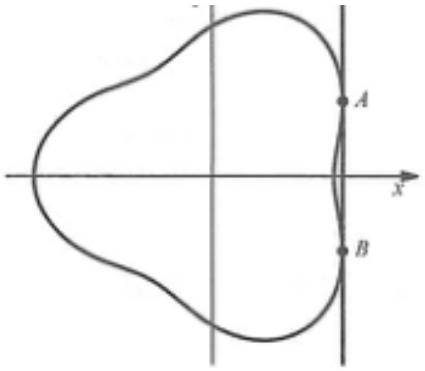

Point $P$ is the point on the curve which has a tangent which is parallel to the initial line.

a) Show that the argument of the point $P$ is given by $\sin\theta = \frac{1}{\sqrt{5}}$.

b) Find the equation of the tangent line in its Cartesian form.

c) Find the area which is bounded by the tangent line going through $P$, the curve and the initial line. Give your answer to three significant figures.

**Question 15.** The diagram shows a sketch of the curve $C$ given by $r = 6\sqrt{3} - 2\cos 3\theta$, $0 \leq \theta < 2\pi$.




The line $L$ is a vertical tangent to the curve which passes through the points $A$ and $B$.

a) Verify that the argument of point $A$ is $\dfrac{\pi}{6}$.

b) Show that the area of the region bounded by $C$ and $L$ as shaded in the diagram is $\left(35\sqrt{3} - \dfrac{55\pi}{3}\right)$ units$^2$.

**Question 16.** For each of the following curves, by using Calculus, find the value of $\theta$ for which the curve is furthest away from the pole and the corresponding distance from the pole:

a) $r = 3\cos\theta$ for $0 \leq \theta \leq \dfrac{\pi}{2}$

b) $r = 2 + \sin\theta$ for $0 \leq \theta \leq \pi$

c) $r = 5\cos 2\theta$ for $0 \leq \theta \leq \dfrac{\pi}{4}$

d) $r = \ln(1 + \sin\theta)$ for $0 \leq \theta < \pi$

**Question 17.** A curve $C$ is given by $r = 4\cos(2\theta)\cosh\left(\frac{\theta}{2}\right)$, $\frac{\pi}{4} \leq \theta \leq \frac{3\pi}{4}$. The point $P$ lies on the curve at the furthest point at angle $\phi$.

a) Explain why $C$ consists of a single loop.

b) Show that $\phi$ satisfies the equation $\phi = \ln\left(\frac{1+4\tan 2\phi}{1-4\tan 2\phi}\right)$.

c) Given that $\phi = 1.65488$ correct to six significant figures, find the distance from $P$ to the pole, giving your answer to four significant figures.

**Question 18.** Let $r = r(\theta)$ be a differentiable function. Given that a curve may be defined parametrically by $x = r\cos\theta$ and $y = r\sin\theta$, show that

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta}\sin\theta + r\cos\theta}{\frac{dr}{d\theta}\cos\theta - r\sin\theta}.$$

**Question 19.** For each of the following polar curves, find the gradient of the tangent line at the given value of $\theta$. Give your answers in exact form.

a) $r = 2\theta$ for $\theta = \dfrac{\pi}{2}$

b) $r = 2\cos\theta$ for $\theta = \dfrac{\pi}{6}$

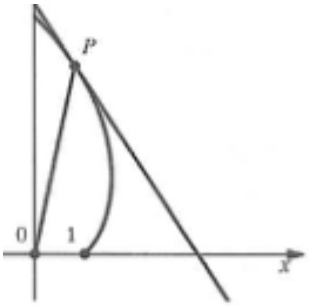
c) $r = 1 + 2\sin\theta$ for $\theta = \dfrac{3\pi}{4}$

**Question 20.** A curve $C$ is given by $r = \theta\sin\theta$ for $0 \leq \theta < \pi$.

a) Show that $\dfrac{dy}{dx} = \dfrac{2\theta\sin 2\theta + 2\sin^2\theta}{2\theta\cos 2\theta + \sin 2\theta}$.

b) Hence, show that this curve consists of a single loop.

**Question 21.** A curve $C$ has a polar equation $r = e^{k\theta}$ for $0 < \theta \leq \frac{\pi}{2}$, where $k$ is a positive constant.




at point $P$ on curve $C$ is $L$
radient of the tangent line $L$

A tangent line at point $P$ on curve $C$ is $L$.

a) Find the gradient of the tangent line $L$.

b) Show that the angle, $\phi$, between the line $OP$ and $L$ satisfies $\tan\phi = \frac{1}{k}$.

**Question 22.** Show that for the curve $r = \sin 2\theta$, the gradients $\frac{dy}{dx}$ at points equidistant from $\theta = \frac{\pi}{4}$ are symmetrically related.